

EXPERIMENT NO: 20

PROGRAM TO READ THE SWITCH AND DISPLAY IN THE LEDs USING LPC2148 KIT

AIM

To write and execute C program to read the switch and display in the LEDs using LPC2148 kit.

OBJECTIVES

- Configure the LPC2148 Port 0 pins used for the LEDs as general-purpose I/O.
- Read the switch input status from the port using IOPIN0.
- Display the switch bit pattern on LEDs P0.15-P0.08.

APPARATUS / SOFTWARE REQUIRED

- Keil uVision5 Software
- Philips Flash Programmer
- LPC 2148 kit

PROGRAM

```
#include "lpc214x.h"

int main(void)
{
    unsigned int sw_sts;

    IODIR0 = 0x0000ff00;    //Configure Port0
    PINSEL0 = 0;           //Configure Port0 as General Purpose IO

    while(1)
    {
        sw_sts = IOPIN0;
        IOSET0 = 0x0000ff00;    //Set P0.15-P0.8 to '1'
        IOCLR0 = sw_sts >> 8;    //Set P0.15-P0.8 to '0'
    }
}
```

OUTPUT

LEDs P0.15-P0.08 displayed the bits entered in the switches

RESULT

Thus C program was written to read the switch and display in the LEDs using LPC2148 kit.